Ali Alavi

Ph.D. student, Computer Science, The Ohio State University · attention decoding, speech, multimodal & video-language models
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Education

The Ohio State University — Columbus, OH
Ph.D. in Computer Science — Sep. 2022 – Present
Adviser: Dr. Donald S. Williamson. Research: auditory attention decoding, multimodal EEG and behavioral sensing, speech quality assessment, neural audio coding.

Indiana University — Bloomington, IN
Ph.D. in Computer Science (transferred to OSU) — Jan. 2022 – Sep. 2022

University of Tehran — Tehran, Iran
M.Sc. in Communication Engineering — Sep. 2016 – Oct. 2020
Thesis: Monte Carlo simulation of polarization-sensitive optical coherence tomography using parallel programming.

Ferdowsi University of Mashhad — Mashhad, Iran
B.Sc. in Electrical and Electronics Engineering — Sep. 2012 – Mar. 2016
Capstone: LNA compensator with LabVIEW and Xmega microcontrollers.

Research Experience

The Ohio State University — Columbus, OH
Graduate Research Associate / GTA — Sep. 2022 – Present

- **Merit** — multimodal attention decoding with per-modality credit — arXiv:2610.17507
  Decoder over EEG, eye tracking, head motion and scene video, with a Shapley-value test of how much accuracy each stream earns. **0.82** / **0.74** four-way accuracy within / across listeners (chance 0.25), above the published benchmark; showed behavioral streams almost halve the EEG's share of accuracy.
- **NEUROTOKEN** — joint source and direction AAD with envelope decoding — arXiv:2610.00397
  One EEG network, three heads; a flow-matching head scores talkers by likelihood ratio: **+9–16%** 5 s source-AAD over the best non-generative baseline (KUL, DTU, NJU), ∼3× lower subject variance. Showed 95–97% direction-AAD results drop 17–45% under trial-disjoint splits.
- **MAESTRO** — open multimodal auditory-attention corpus — arXiv:2609.31898
  Co-developed the first AAD corpus with simultaneous EEG, gaze, pupillometry, egocentric video and head IMU (four talkers, several SNRs); designed and built the EEG + eye-tracking recording setup: IRB protocol, stimulus delivery, hardware synchronization, participant sessions. Public on Hugging Face.
- **Speech-quality metric hackability** — corrected protocol and benchmark — arXiv:2610.10899
  Corrected how reward-hacking of quality predictors is measured (changes results up to 4×); NISQA is hacked on **90%** of utterances vs. SSL-MOS 21%, DNSMOS 14%, UTMOS 6%; audited attack-detect-patch hardening.
- **CADENZA** — content-adaptive neural audio codec — *Interspeech 2026 submission*
  Sub-kilobit speech compression via variable-rate finite scalar quantization, conditional flow matching and a streaming Conformer.
- **Contrastive AAD** — self-supervised EEG–speech alignment — arXiv:2410.18395
  Aligned EEG and attended-speech representations, then fine-tuned for attention classification; state of the art on the validation set at publication.

Indiana University — Bloomington, IN
Graduate Research Assistant — Jan. 2022 – Sep. 2022

- Started research on EEG-guided speech enhancement with deep learning under Dr. Donald S. Williamson.

Competitions & Challenge Reports

- **Silver Medal, 145th place** — BirdCLEF+ 2026, Kaggle. — certificate
- **2nd place, both submitted proposals** — Obesity ML Challenge Part 3 (peer review), CrunchDAO, 2026. — results
- **TimeLogic Challenge @ CVPR 2026** — *TLG: Temporal-Logic Grounding for Video Question Answering via Source-Annotation Reconstruction and Category-Targeted Reasoning.* arXiv:2606.01591
  71.37% test accuracy (+24.5 over a VLM baseline), within 3 points of the leaderboard top.
- **VRR Challenge @ CVPR 2026** — *Perception First: A Frontier Native-Video Model with Self-Consistency for Implicit Video Question Answering.* arXiv:2606.01485
  81.18% on the hidden test split, above the prior best (80.85%), training-free.
- **CoVR-R Challenge, VidLLMs Workshop @ CVPR 2026** — *Reason, Retrieve, Re-rank: A Zero-Shot Reasoning-Aware Framework for Composed Video Retrieval.* arXiv:2606.00910
  91.9% R@1 and 98.2% R@10 on the test set with frozen models only (Qwen3-VL, SigLIP-2).

Publications & Preprints

[1] **A. Alavi**, D. S. Williamson. "Merit: Making every modality earn its contribution in multimodal attention decoding." arXiv:2610.17507, 2026.

[2] **A. Alavi**, D. S. Williamson. "How Hackable Is Your Speech Quality Metric? A Corrected Protocol, a Benchmark, and What Patching Buys." arXiv:2610.10899, 2026.

[3] **A. Alavi**, D. S. Williamson. "NEUROTOKEN: Joint Source and Directional AAD with Envelope Decoding via Conditional Flow Matching." arXiv:2610.00397, 2026.

[4] K. M. N. Hassan, **A. Alavi**, D. S. Williamson. "MAESTRO: a Multimodal Auditory-attention Egocentric Speech-TRacking Open corpus." *Submitted to IEEE Trans. Audio, Speech, and Language Processing*; arXiv:2609.31898, 2026.

[5] **A. Alavi**, D. S. Williamson. "CADENZA: Content-Adaptive Encoding with Zero-Collapse Quantization for Sub-Kilobit Neural Audio Compression." *Submitted to Interspeech*, 2026.

[6] **S. A. Alavi Bajestan**, M. Pitt, D. S. Williamson. "A contrastive-learning approach for auditory attention detection." arXiv:2410.18395, 2024.

[7] **A. Alavi**. "Neural network for estimation of optical characteristics of optically active and turbid scattering media." arXiv:2011.06934, 2020.

Industry Experience

BNtelligence Columbus, OH
Computer Vision Engineer Jul. 2025 – Aug. 2025

- Built an end-to-end defect-detection pipeline: data collection and annotation, classification and semantic-segmentation models, and a real-time video inference API serving production traffic.

AI-bridge Hamburg, Germany
Deep Learning Consultant Oct. 2019 – Mar. 2020

- Shipped production deep-learning applications (image enhancement, body-part segmentation, super-resolution) in TensorFlow and PyTorch.

Payafanavaran Mashhad, Iran
Software Engineer Jan. 2019 – Aug. 2019

- Developed a C# application for 3D surface reconstruction from structured-light scanning data.

Tarashe Sanat-e Pishro Mashhad, Iran
Hardware Design Engineer Jul. 2016 – Sep. 2016

- Designed a high-speed FPGA board (DDR3 SDRAM, Ethernet, peripherals) for industrial data acquisition.

Technical Skills

Research auditory attention decoding, EEG/BCI, multimodal fusion and attribution, speech quality assessment, neural audio coding, flow matching and diffusion, self-supervised and contrastive learning, video-language models, RL for LLMs (GRPO, RLVR)

Languages Python, C/C++, C#, CUDA, MATLAB, Java, JavaScript

ML / Infra PyTorch, Lightning, TensorFlow, scikit-learn, Hugging Face, Qwen-VL, SigLIP-2, Gemini API, Git, Linux, Docker, HPC/Slurm, W&B, Hydra, LaTeX, Lean 4

Hardware EEG and eye-tracking acquisition and sync, FPGA (VHDL/Verilog), ARM Cortex-M, LabVIEW, Cadence

Honors & Service

- CCBS Summer Research Award, The Ohio State University, 2023 · 3rd Place, CMDC Hackathon
- Reviewer, EACL 2021 AfricaNLP Workshop · Student Volunteer, ICLR 2021, ICML 2021, ICEE 2015 · Coordinator, Entrepreneurship Center, Ferdowsi University of Mashhad, 2015–2016